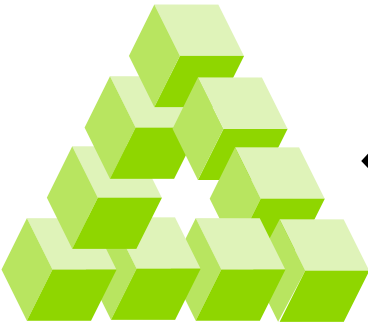




Just an illusion

This mind-bending shape is called a Penrose triangle. The pattern of shade tricks the human brain into seeing a 3D triangle, each corner of which is 90° (a right angle) – a mathematical impossibility.



Can you find two skulls hidden in the painting on the left?



A hidden skull

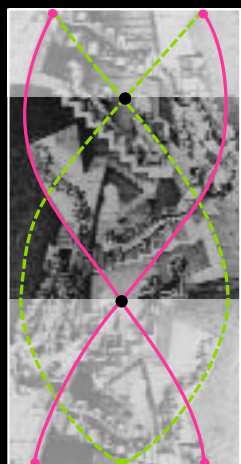
The Ambassadors, painted in 1533 by Holbein, has a peculiar smear at the bottom. If you hold the page close to your eye and look at the smear from lower left or upper right, you'll see it's a skull. Holbein used perspective geometry to draw this skull, but its meaning is a mystery. There is also a second, much tinier, skull on one of the men's caps.



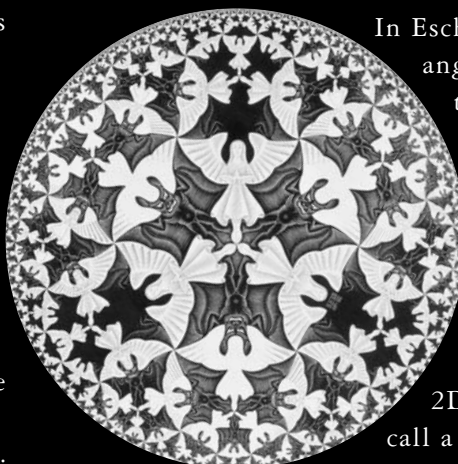
3D art

These days, people use computers to create 3D images. The picture above is a stereogram. If you look through it and move it slowly forward or back, you'll see sweets magically floating in midair.

to create optical effects,
such as the *illusion* of three dimensions



Escher's *House of Stairs* looks impossible because he gave it two vanishing points and made parallel lines run towards and around them in curves. Each of the creatures (which Escher called "curl-ups") belongs to one vanishing point or the other. The top of the picture is a repetition of the bottom, so the scene might conceivably carry on forever.



In Escher's *Circle Limit IV*, angels and demons form a tessellating pattern, with the spaces between one cleverly forming the shapes of the other. The pattern shrinks towards the edge, seeming to continue to infinity. Escher created this picture to represent an impossible 2D surface that mathematicians call a hyperbolic plane.